Price after second discount = 75 – 15 = ₹60

$\therefore$ Net selling price = ₹60

$\therefore$ Single discount in percentage equivalent to two given successive discounts = (100–60)% = 40%.



Note

- If there are 2 successive discounts of a % and b % respectively, then
  $\text{S.P} = \left(1-\frac{a}{100}\right)\left(1-\frac{b}{100}\right)\times \text{M.P.}$
- Single discount in % equivalent to 3 successive discounts of a %, b % and c % respectively
  $= \left\{1-\left(1-\frac{a}{100}\right)\left(1-\frac{b}{100}\right)\left(1-\frac{c}{100}\right)\right\}\times 100\%$.

(Use this formula for the Example 4.11 and check the answer).

Example 4.12

A water heater is sold by a trader for ₹10502 including GST at 18%. Find the marked price of the water heater and GST.

Solution:

Let the marked price be ₹ x.

$$\text{Now, } x+\frac{18x}{100} = 10502$$

$$\frac{118x}{100} = 10502$$

$\therefore$ Marked price, x = ₹8900

GST at 18% = ₹10502 – ₹8900 = ₹1602

(or)

$$= 8900 \times \frac{18}{100} = ₹1602$$

Example 4.13

A family went to a hotel and spent ₹350 for food and paid extra 5% as GST. Calculate the CGST and SGST.

Solution:

Cost of food = ₹350

Extra 5% paid as GST is equally shared by the Central and the State Governments at 2.5% each.

$$\therefore \text{CGST} = \text{SGST} = 350 \times \frac{2.5}{100} = ₹8.75$$

Exercise 4.2

1. Fill in the blanks:

(i) Loss or gain percentage is always calculated on the_____.

(ii) A mobile phone is sold for ₹8400 at a gain of 20%. The cost price of the mobile phone is_____.

04_8th_Maths_Life Mathematics_Chap 4.indd 130 2/12/2020 12:14:10 PM